

Cloud-Networking -Assignment-1

Q1. Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).

Ans:

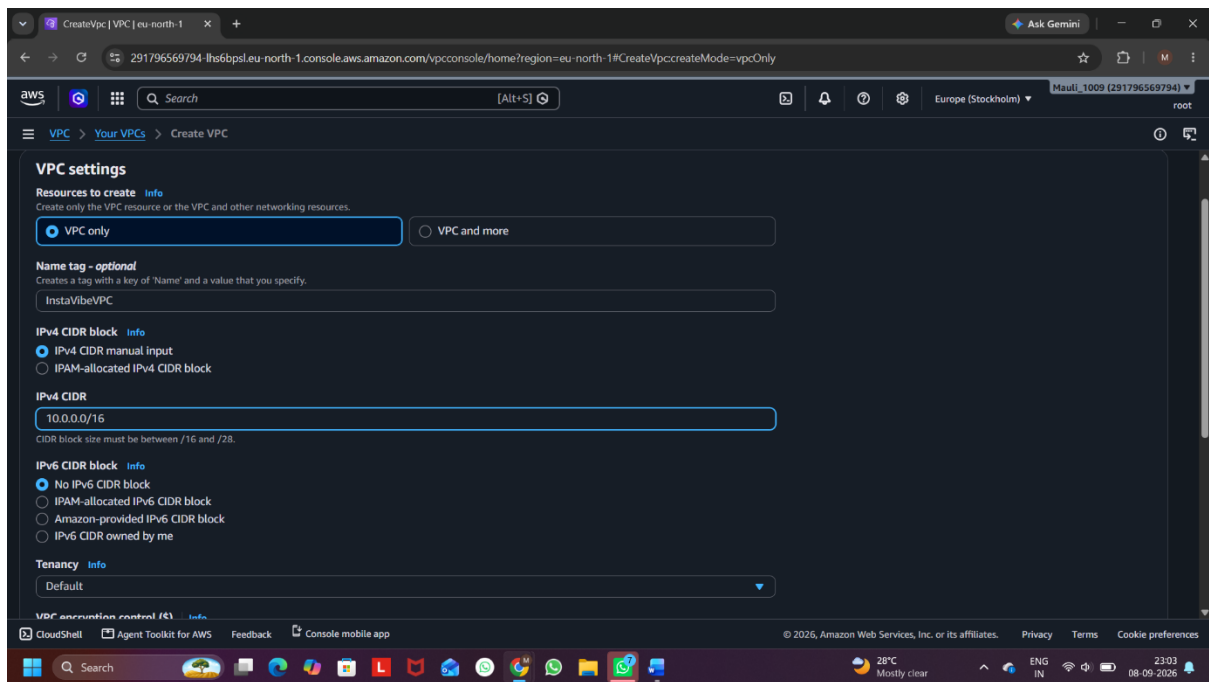


Figure 1: VPC creation settings with the name and IPv4 CIDR block.

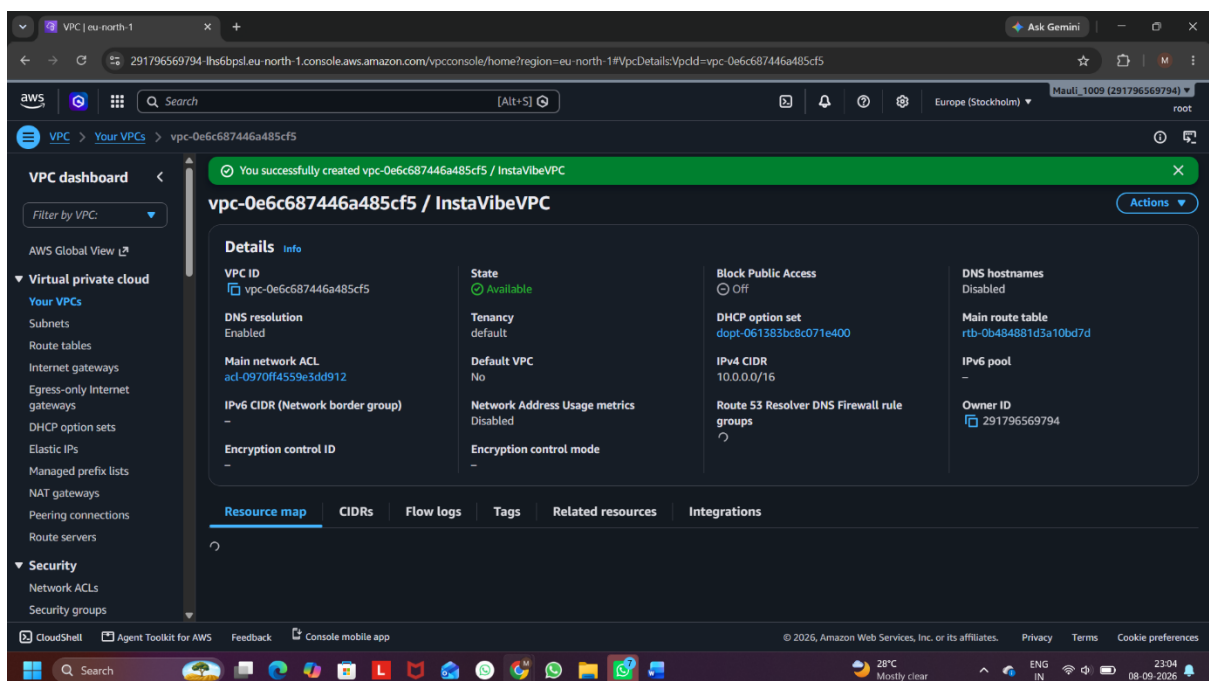


Figure 2: Successfully created InstaVibeVPC.

Q2. Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.1.0/24) for hosting a web server and one private subnet (e.g., 10.0.2.0/24) for backend services. Hint: Name your subnets as 'PublicSubnet-Zomato' and 'PrivateSubnet-Zomato' for clarity.

Ans:

Inside the InstaVibeVPC, I created two subnets. The first subnet was created as PublicSubnet-Zomato with the CIDR block 10.0.1.0/24 for hosting the web server. The second subnet was created as PrivateSubnet-Zomato with the CIDR block 10.0.2.0/24 for backend services.

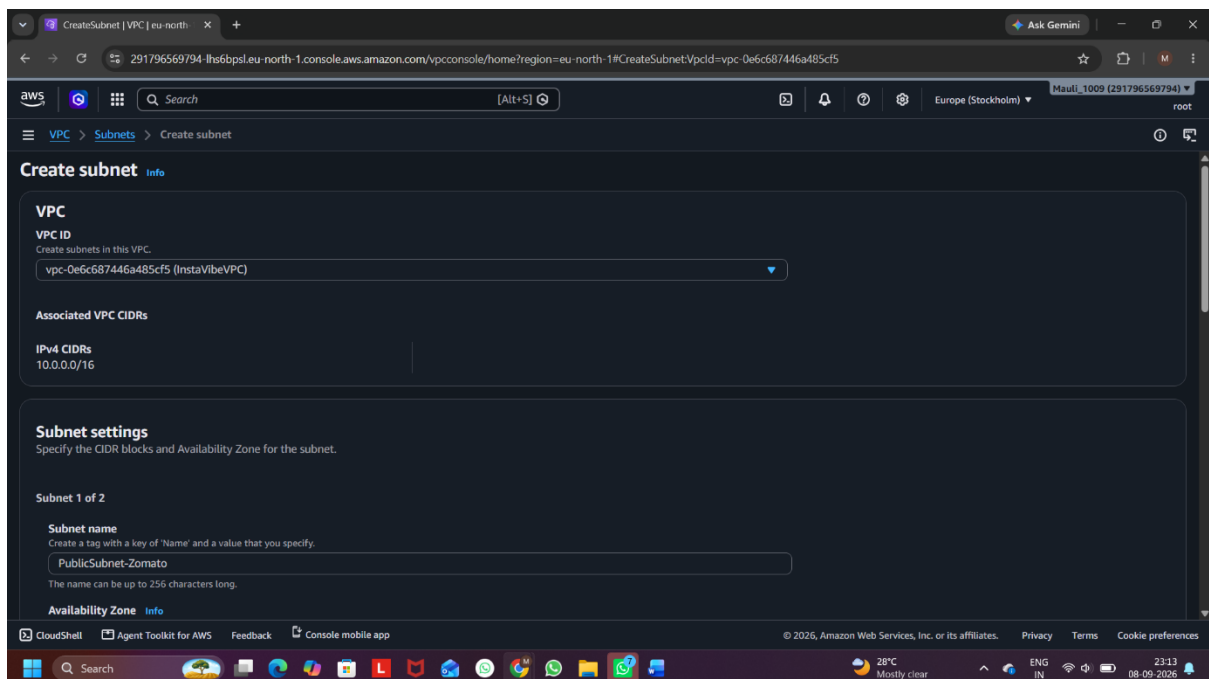


Figure 1: PublicSubnet-Zomato created successfully.

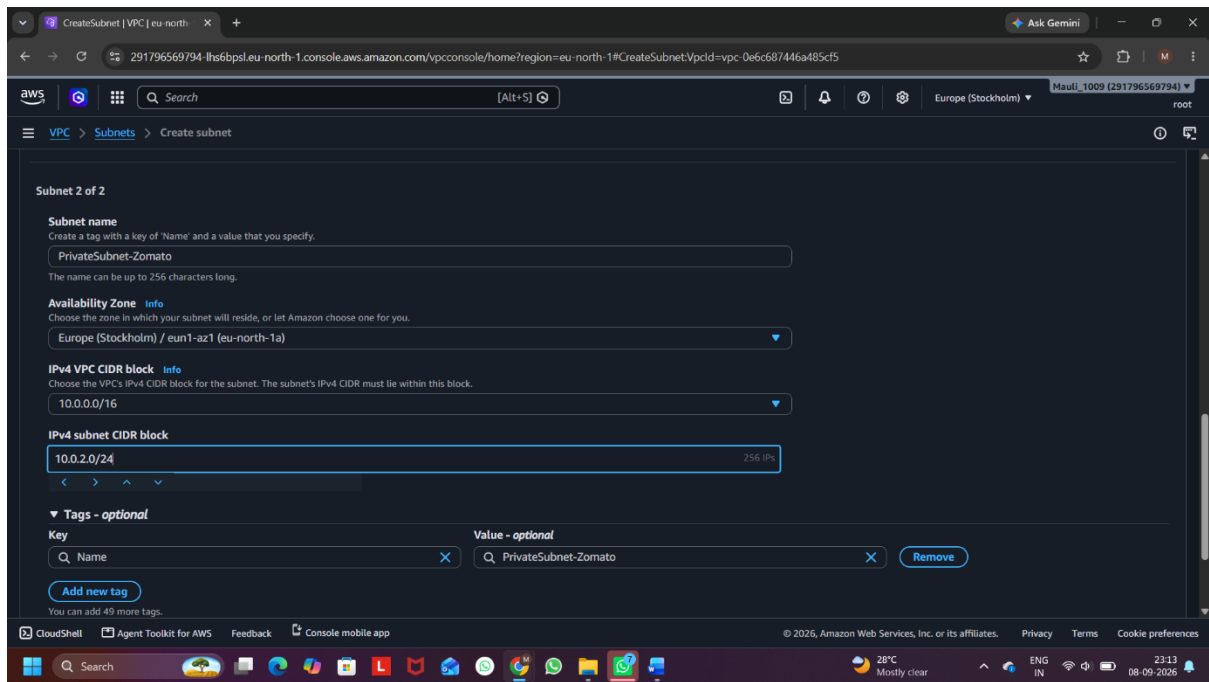


Figure 2: PrivateSubnet-Zomato created successfully.

Q3. Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.

Ans:

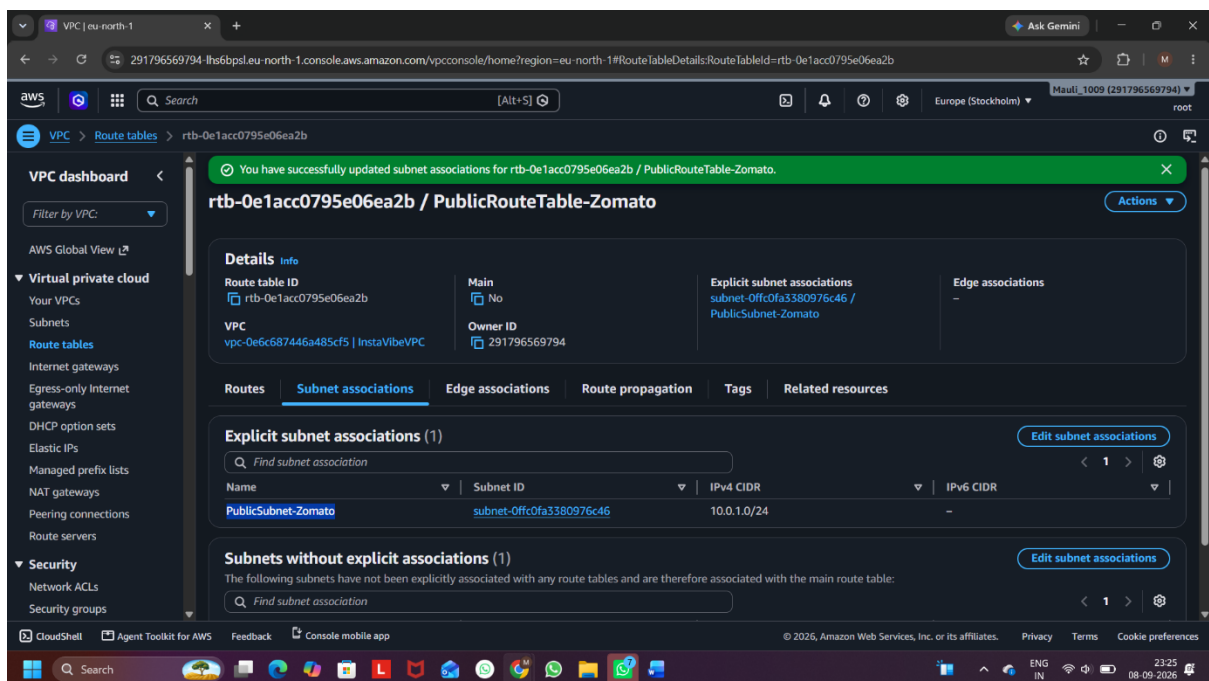


Figure 1: PublicSubnet-Zomato associated with PublicRouteTable-Zomato

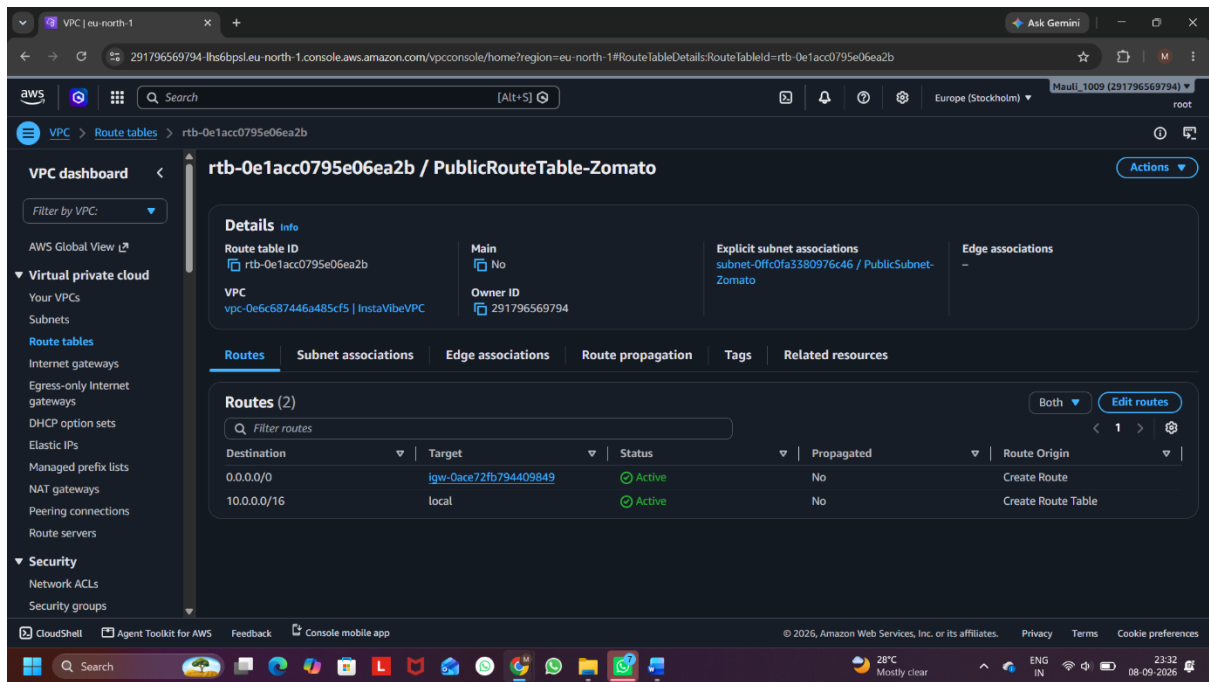


Figure 2: 0.0.0.0/0 → InstaVibe-IGW route

Q4. Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.

Ans:

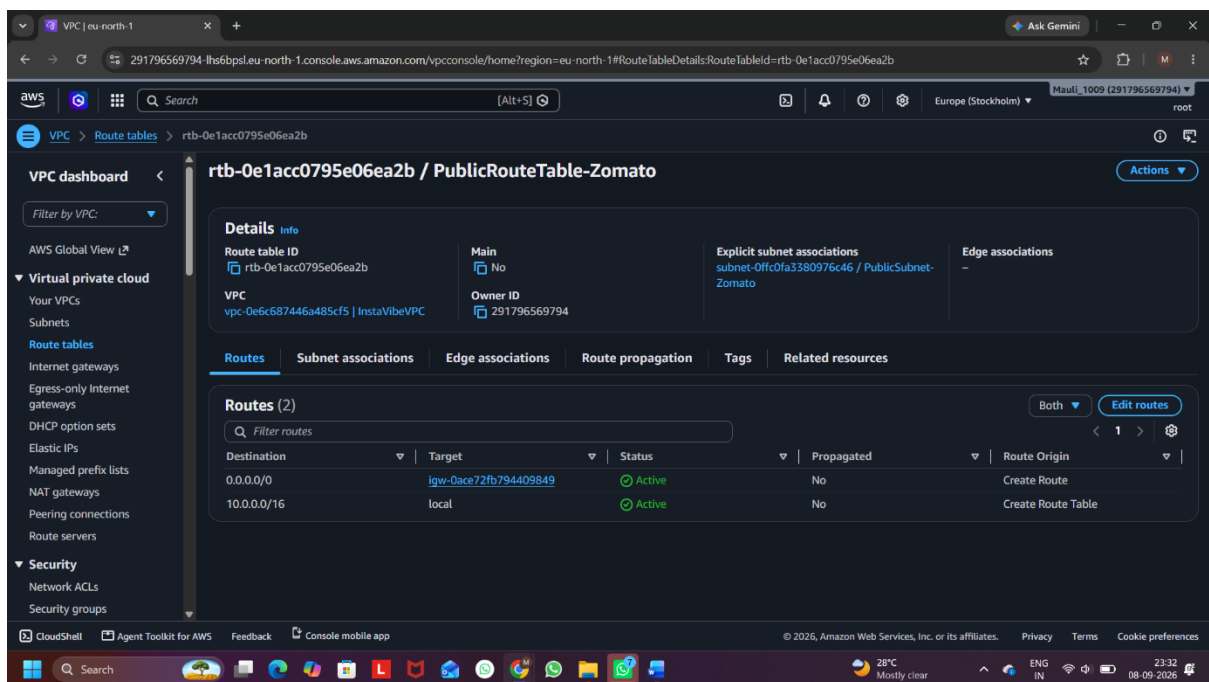


Figure 1: Public Route Table with Internet Gateway Route

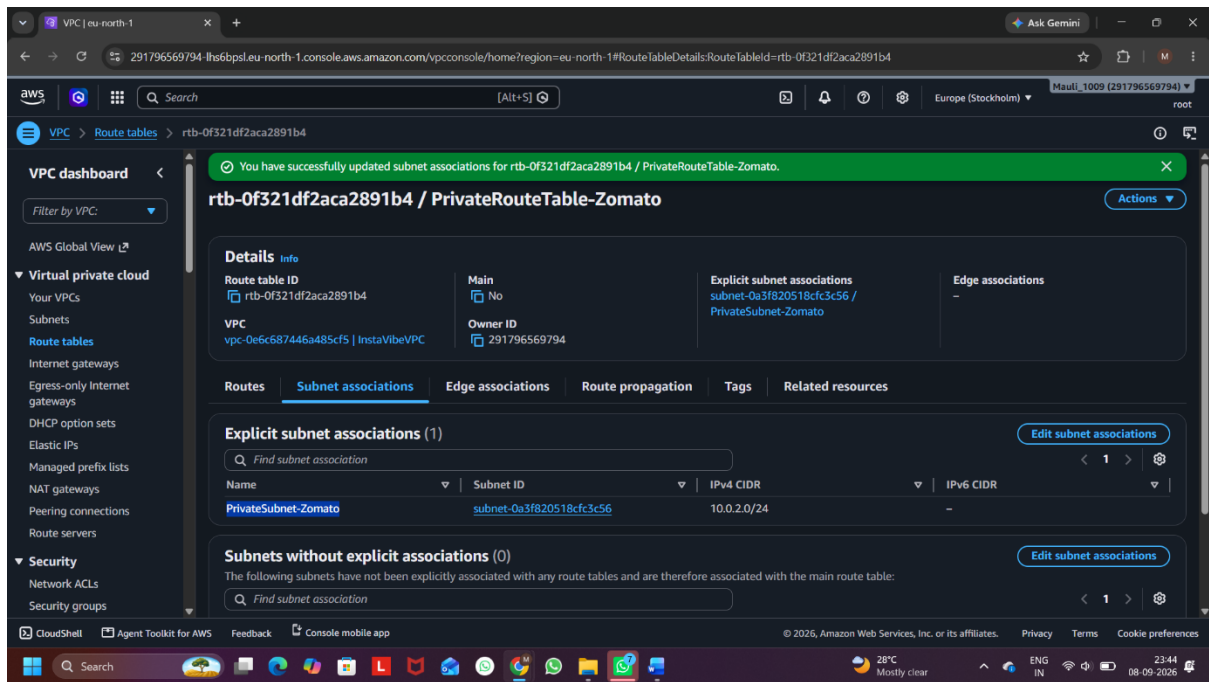


Figure 2: PrivateSubnet-Zomato associated with PrivateRouteTable-Zomato

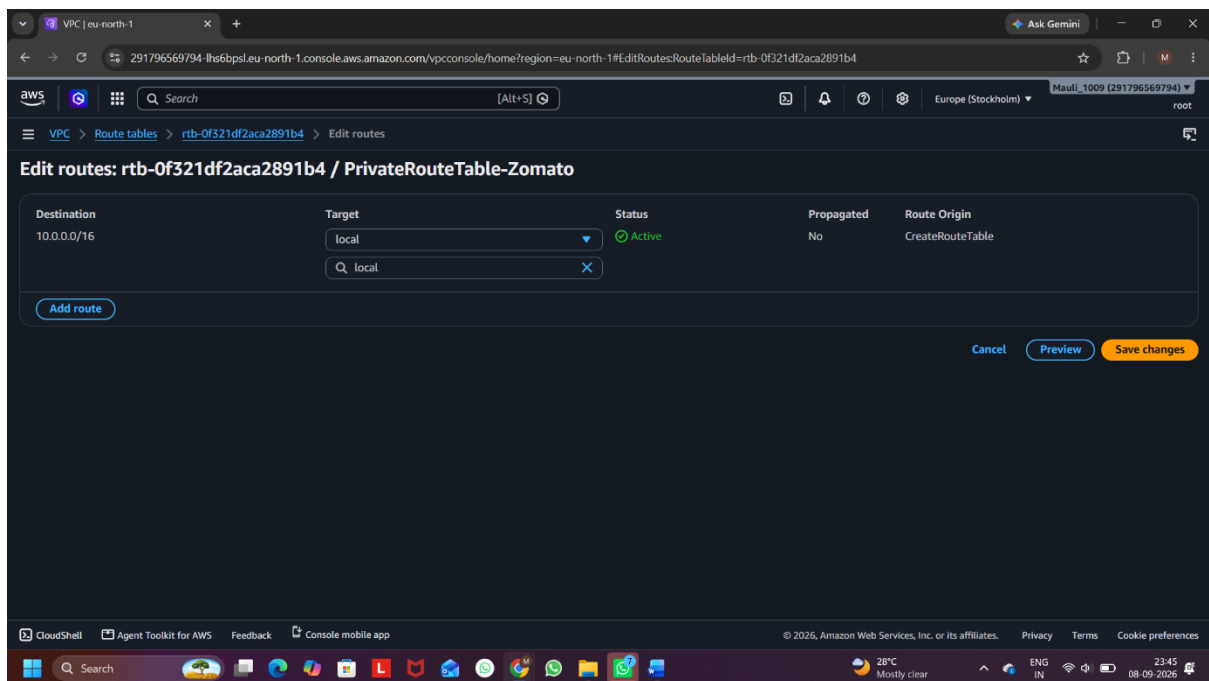


Figure 3: PrivateRouteTable-Zomato routes showing no 0.0.0.0/0 → IGW

Q5. Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.

Ans:

